Gavin Kerrigan

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Research

Florence Nightingale Bicentenary Fellow in Statistical Machine Learning Sep. 2026 -
University of Oxford, Department of Statistics

Postdoctoral Research Assistant in Machine Learning Feb. 2025 - Sep. 2026
University of Oxford, Department of Statistics
RainML Lab (PI: Tom Rainforth)

Graduate Student Researcher Sep. 2019 - Dec. 2024
University of California, Irvine, Department of Computer Science
DataLab (PI: Padhraic Smyth)

Undergraduate Research Assistant Sep. 2018 - June 2019
The Pennsylvania State University, Department of Computer Science

Education

Ph.D., Computer Science Sep. 2019 - Dec. 2024
University of California, Irvine GPA: 4.0/4.0
Thesis: Infinite-Dimensional Generative Models through the Transport of Measure
(Best Ph.D. Dissertation Award, UCI Department of Computer Science)
Advisor: Padhraic Smyth

B.Sc., Mathematics 2015 - 2019
The Pennsylvania State University, Schreyer Honors College GPA: 3.98/4.0
Minor: Physics

Teaching

Stipendiary Lecturer in Statistics Oct. 2026 -
Jesus College, University of Oxford

Instructor
SC4: Advanced Topics in Statistical Machine Learning
University of Oxford Jan. 2027 - Mar. 2027
- Enrollment: TBD

CS 178: Machine Learning and Data Mining
UC Irvine March 2023 - June 2023
- Enrollment: 162

Tutor Jan. 2026 - Mar. 2026
SB2.2: Statistical Machine Learning, University of Oxford

Head Teaching Assistant Sep. 2022 - Dec. 2022

CS 178: Machine Learning and Data Mining, UC Irvine

- Managed a team of 5 graduate TAs in a course with an enrollment of 272
- Developed homework assignments, weekly discussions, exams, and evaluation rubrics

Guest Lecturer

StatML CDT: Bayesian Experimental Design, University of Oxford June 2026

Fundamentals of AI: Modern Statistical Concepts, University of Oxford Nov. 2025, Nov. 2026

CS 274E: Deep Generative Models, UC Irvine Nov. 2022, Nov. 2023, Nov. 2024

Freelance Private Tutor Jul. 2019 - Dec. 2024

Machine learning, mathematics, and statistics

- One-on-one, individualized instruction to students ranging from secondary school to Ph.D. level
- Perfect (5.0/5.0) rating across more than 300 hours of tutoring and 200 reviews

Mathematics Tutor and Peer Coordinator Jan. 2017 - June 2019

Penn State Learning

- Tutored undergraduate mathematics courses in one-on-one and small group settings
- Trained, supervised, and oversaw hiring of a team of more than 60 undergraduate tutors

Publications

Google Scholar: https://scholar.google.com/citations?user=2F2XCy8AAAAJ

** denotes joint authorship*

Conference

1. A. Phillips, **G. Kerrigan**, T. Rainforth, 'Bayesian Experimental Design via Score Matching,' *Forty-Second Annual Conference on Uncertainty in Artificial Intelligence (UAI 2026)*, to appear, August 2026.
2. **G. Kerrigan**, K. Nelson, P. Smyth, 'EventFlow: Forecasting Temporal Point Processes with Flow Matching,' *Proceedings of the 29th International Conference on AI and Statistics (AISTATS 2026)*, May 2026 **(Best Paper Award)**.
3. **G. Kerrigan**, C. Naesseth, T. Rainforth, 'A Geometric Approach to Optimal Experimental Design,' *Proceedings of the 29th International Conference on AI and Statistics (AISTATS 2026)*, May 2026.
4. A. Saravanan, R. Knowles, **G. Kerrigan**, T. Rainforth, 'Scaling Bayesian Experimental Design to High-Dimensions with Information-Guided Diffusion', *International Conference on Learning Representations (ICLR 2026)*, pp. 131048-131073, April 2026.
5. A. Mammadov, J. Yao, J. Berner, **G. Kerrigan**, J. C. Ye, K. Azizzadenesheli, A. Anandkumar, 'Guided Diffusion Sampling on Function Spaces with Applications to PDEs,' *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, pp. 127057-127094 December 2025.
6. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Dynamic Conditional Optimal Transport through Simulation-Free Flows,' *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pp.93602–93642, December 2024.

7. P. Srivastava, R. Yang, **G. Kerrigan**, G. Dresdner, J. McGibbon, C. Bretherton, S. Mandt, 'Precipitation Downscaling with Spatiotemporal Video Diffusion,' *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pp. 56374-56400, December 2024.
8. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Functional Flow Matching,' *Proceedings of the 27th International Conference on AI and Statistics (AISTATS)*, Proceedings of Machine Learning Research, PMLR 238:3934-3942, April 2024 **(Oral, Outstanding Student Paper Highlight award (7/2181))**.
9. **G. Kerrigan**, J. Ley, P. Smyth, 'Diffusion Generative Models in Infinite Dimensions,' *Proceedings of the 26th International Conference on AI and Statistics (AISTATS)*, Proceedings of Machine Learning Research, PMLR 206:9538–9563, April 2023.
10. **G. Kerrigan**, P. Smyth, and M. Steyvers, 'Combining Human Predictions with Model Probabilities via Confusion Matrices and Calibration,' *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pp.4421–4434, December 2021.

Journal

11. C. Guilloteau, **G. Kerrigan,** K. Nelson, G. Migliorini, P. Smyth, R. Li, E. Foufoula-Georgiou, 'A Generative Diffusion Model for Probabilistic Ensembles of Precipitation Maps Conditioned on Multisensor Satellite Observations,' *IEEE Transactions on Geoscience and Remote Sensing*, doi:10.1109/TRGS.2025.3548518, March 2025.
12. M. Steyvers, H. Tejeda, **G. Kerrigan**, P. Smyth, 'Bayesian Modeling of Human-AI Complementarity,' *Proceedings of the National Academy of Sciences*, 19(11):1-7, March 2022.

Workshop

13. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Functional Flow Matching,' *NeurIPS Workshop on Diffusion Models*, 2023.
14. **G. Kerrigan**[*], D. Slack[*], J. Tuyls[*], 'Differentially Private Language Models Benefit from Public Pre-Training,' *PrivateNLP: EMNLP Workshop on Privacy and Natural Language Processing*, pp.39-45, 2020.

Under Review

15. **G. Kerrigan**, M. Hedman, S. Iqbal, T. Rainforth, 'Robust Experimental Design Should Target Actions, Not Uncertainty,' *under review*, October 2026.
16. **G. Kerrigan**, M. Hedman, S. Iqbal, T. Rainforth, 'TestBED: An Open-Source Toolkit for Bayesian Experimental Design,' *under review*, September 2026.
17. A. Forster, **G. Kerrigan**, D. R. Ivanova, T. Rainforth, 'Improving Robustness to Model Misspecification in Bayesian Experimental Design,' *under review*, January 2026.

Funding and Awards

Funding Sep. 2026

Joint principal investigator, UKRI AIRR Gateway Project 0261-8037-7461-1

- "TestBED: A Framework for Benchmarking Deep Adaptive Data Gathering"

- 15,000 GH200 GPU hours awarded in competitive supercomputing grant

Research

Best Ph.D. Dissertation Award, UC Irvine Department of Computer Science	2026
Best Paper Award, AISTATS	2026
Outstanding Student Paper Highlight Award, AISTATS	2024

University of California, Irvine

Hasso Plattner Institute Fellowship	2021 - 2024
Graduate Dean's Recruitment Fellowship	2019

The Pennsylvania State University

Steven and Sherry McCrystal Mathematics Award	2019
Mary Lister McCammon Scholarship	2017, 2018
Phi Beta Kappa Honors Society	2017
Hay Memorial Scholarship	2016
Mahle Trustee Scholarship in Science	2015

Invited Talks

Model Misspecification and Robust Experimental Design

Computational Statistics and Machine Learning Seminar, Lancaster University	Oct. 2026
Statistics and Data Science Seminar, University of Birmingham	Oct. 2026

Optimal Transport and Optimal Experimental Design

CFE-CMStatistics	December 2026
Computational Methods in Applied Mathematics	July 2026

Generative Models and the Transport of Measure

Anandkumar Group, Caltech	August 2024
University of Southern California (USC)	October 2024
Yu Group, University of California San Diego (UCSD)	October 2024

Diffusion Generative Models: Methods and Applications

NASA Jet Propulsion Laboratory (JPL)	June 2024

Functional Flow Matching

The 27th International Conference on AI and Statistics (AISTATS)	May 2024

Deep Generative Models in Infinite-Dimensional Spaces

Center for Machine Learning Seminar, UC Irvine	Jan. 2024
Probability and Combinatorics Seminar, UC Irvine	Nov. 2023
AMLab Seminar, University of Amsterdam	Nov. 2023

A Tutorial on Diffusion Generative Models

Hasso Plattner Institute	Sep. 2023

Supervision

MSc. in Statistical Science Dissertation Supervisor

University of Oxford, Department of Statistics

Dominik Baus, “High-Dimensional Sampling Algorithms utilizing Wasserstein Gradient Flows” 2026

Linxuan Wang, “Generative Modelling via Drifting on Manifolds” 2026

Service

Computational Statistics and Machine Learning Seminars Co-Organiser Oct. 2026 -

University of Oxford, Department of Statistics

The Network Co-Organiser Sep. 2026 -

University of Oxford, Department of Statistics

Bayesian Experimental Design & Active Learning (BEDAL) Seminar Co-Organiser Sep. 2026 -

Ellison Scholars Application Assessor

Ellison Institute of Technology, Oxford Aug. 2026

Facilities and Safety Committee Florence Nightingale Representative

University of Oxford, Department of Statistics Sep. 2026 -

Departmental Committee Postdoctoral Representative

University of Oxford, Department of Statistics Nov. 2025 - Sep. 2026

DPhil Admissions Interviewer

University of Oxford, Department of Statistics Jan. 2026

Student Success Mentor

The Mentor Collective, UC Irvine Next Gen Pathways 2024

Workflow Chair

AISTATS 2024

- Oversaw and managed the peer review workflow for an ML conference with 2,000+ submissions, ensuring timely handling of reviews and decisions
- Coordinated reviewer assignments, monitored review progress, and resolved workflow issues in collaboration with program chairs and area chairs
- Managed conference website and handled the majority of conference-related correspondence with authors, reviewers, and committee members

Session Chair

AISTATS 2024

Area Chair

ICML, NeurIPS, ICLR 2026

ICLR 2025

AI Faculty Search Committee, Student Member

UC Irvine 2021 - 2023

Machine Learning Hackathon Mentor

UC Irvine 2021

Academic Reviewing

UAI, ProbML, International Statistical Review, JMLR, AISTATS 2026

JMLR, ICML, NeurIPS, AAAI, AABI, AISTATS, TAG-DS, The American Statistician	2025
JMLR, ICLR, CVPR, ICML, UAI, AAAI, AISTATS, NeurReps Workshop	2024
NeurIPS, AISTATS, Workshop on Deep Generative Models for Health, Workshop on Topology, Algebra, and Geometry in Data Science, Workshop on Diffusion Models	2023
ICML, Workshop on Geometrical and Topological Representation Learning, Workshop on Human-Machine Collaboration and Teaming	2022
Workshop on Uncertainty in Deep Learning	2020